Tea and toast: systems and energy

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 State the energy forms for the input and output of a kettle.

- Input: electrical energy
- Output: heat (thermal) energy (useful), with some sound

2 Draw a universal system diagram for a toaster.

- Input: electrical energy (or "switch on") → Process: toaster → Output: heat (toasted bread)

3 A toaster weighs 15 N. Calculate the work done carrying it up a staircase with a vertical height of 4.2 m.

- $E_W = F \times d = 15 \times 4.2 = 63$ J

4 Draw a sub-system diagram for an automatic kettle. Include the feedback loop and the system boundary.

- Set temperature (input) → control → output driver → heating element → hot water (output)
- Temperature sensor measuring the water, feeding back to the control
- Dashed system boundary round the sub-systems

5 Describe, with reference to your sub-system diagram, the operation of the kettle.

- SET: the water temperature is set (boiling).
- ACTION: the control switches the heating element on through the driver.
- FEEDBACK: the temperature sensor measures the actual water temperature.
- COMPARE: the control compares the actual and set temperatures.
- ADJUST: when the water boils, the heating element switches off.

6 It takes 58.7 J of work to carry the empty kettle up to the Art department, 7.68 m higher. Calculate the mass of the kettle.

- $F = E_W \div d = 58.7 \div 7.68 = 7.64$ N
- $m = F \div g = 7.64 \div 9.8 = 0.78$ kg

7 Describe how the kettle could be made more energy efficient.

- Insulate the body so less heat escapes; use a sensor to switch off exactly at boiling; fill only with the water needed.